

Problem 1 : Basic Exam, Spring 2022

Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\lim_{N \rightarrow \infty} \int_0^{2\pi} f(x) \sin(Nx) \, dx = 0.$$

Solution. $\forall \varepsilon > 0$. Let $\varepsilon' = \frac{\varepsilon}{4\pi}$. By Weierstrass approximation theorem, $\exists p \in \mathbb{R}[x]$ such that $\|f - p\|_\infty < \varepsilon'$. Note $p' : [0, 2\pi] \rightarrow \mathbb{R}$ and thus $\|p'\|_\infty < \infty$. There exists $M \in \mathbb{N}$ such that $\forall N \geq M$ we have that

$$\begin{aligned} \frac{4\pi\|p'\|_\infty}{N} &< \frac{\varepsilon}{2} \\ \left| \int_0^{2\pi} f(x) \sin(Nx) \, dx \right| &= \left| \int_0^{2\pi} (f(x) - p(x) + p(x)) \sin(Nx) \, dx \right| \\ &\leq \left| \int_0^{2\pi} (f(x) - p(x)) \sin(Nx) \, dx \right| + \left| \int_0^{2\pi} p(x) \sin(Nx) \, dx \right| \\ &< 2\pi\varepsilon' + \left| \frac{1}{N} [p(x) \cos(Nx)]_0^{2\pi} + \frac{1}{N} \int_0^{2\pi} p'(x) \cos(Nx) \, dx \right| \\ &= 2\pi\varepsilon' + \left| \frac{1}{N} [p(0) - p(2\pi)] + \frac{1}{N} \int_0^{2\pi} p'(x) \cos(Nx) \, dx \right| \\ &= 2\pi\varepsilon' + \left| \frac{1}{N} \int_0^{2\pi} (\cos(Nx) - 1) p'(x) \, dx \right| \\ &\leq 2\pi\varepsilon' + \frac{1}{N} \int_0^{2\pi} |\cos(Nx) - 1| \|p'\|_\infty \, dx \end{aligned}$$

Note that $-1 \leq \cos(Nx) \leq 1 \Rightarrow -2 \leq \cos(Nx) - 1 \leq 0 \Rightarrow |\cos(Nx) - 1| \leq 2$. Thus

$$\left| \int_0^{2\pi} f(x) \sin(Nx) \, dx \right| < 2\pi\varepsilon' + \frac{4\pi\|p'\|_\infty}{N} \leq 2\pi\varepsilon' + \frac{4\pi\|p'\|_\infty}{M} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus we have

$$\lim_{N \rightarrow \infty} \int_0^{2\pi} f(x) \sin(Nx) \, dx = 0$$

Problem 2 : Basic Exam, Fall 2007

Let $f \in C^2(\mathbb{R})$ be such that f'' is a bounded function on $\mathbb{R}$.

(a) Let $A > 0$. Show that

$$\left| \int_{-A}^A f(x) \, dx - 2Af(0) \right| \leq \frac{A^3}{3} \|f''\|,$$

where $\|f''\| = \sup_{x \in \mathbb{R}} |f''(x)|$.

(b) Let $[a, b] \subseteq \mathbb{R}$ be a compact interval. Show that there exists a constant $C > 0$ depending on a, b , and $\|f''\|$, such that for all $n = 1, 2, \dots$,

$$\left| \int_a^b f(x) \, dx - \frac{b-a}{n} \sum_{k=1}^n f\left(a + \frac{b-a}{2n}(2k-1)\right) \right| \leq \frac{C}{n^2}.$$

Solution. (a): $\forall x \in [-A, A]$ there exists a $c \in [0, x] \cup [x, 0]$ where

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(c)x^2$$

$$f(x) - f(0) - f'(0)x = \frac{1}{2}f''(c)x^2$$

$$\int_{-A}^A [f(x) - f(0) - f'(0)x] \, dx = \frac{1}{2}f''(c) \int_{-A}^A x^2 \, dx$$

$$\int_{-A}^A f(x) \, dx - 2f(0)A = \frac{1}{6}f''(c)2A^3$$

$$\left| \int_{-A}^A f(x) \, dx - 2f(0)A \right| \leq \frac{1}{3}\|f''\|_\infty A^3$$

(b): Note the following chain of equalities.

$$\begin{aligned} \int_a^b f(x) \, dx &= \sum_{k=1}^n \int_0^{\frac{b-a}{n}} f\left(x + \frac{k-1}{n}(b-a) + a\right) \, dx \\ &= \sum_{k=1}^n \int_{-\frac{b-a}{2n}}^{\frac{b-a}{2n}} f\left(x + a + \frac{b-a}{2n}(2k-1)\right) \, dx \end{aligned}$$

Let $t_n(k) = a + \frac{b-a}{2n}(2k-1)$ and let $A = \frac{b-a}{2n}$.

We can rewrite the difference as:

$$\begin{aligned} \int_a^b f(x) \, dx - \frac{b-a}{n} \sum_{k=1}^n f(t_n(k)) &= \sum_{k=1}^n \int_{-A}^A f(x + t_n(k)) \, dx - \sum_{k=1}^n 2Af(t_n(k)) \\ &= \sum_{k=1}^n \left[\int_{-A}^A f(x + t_n(k)) \, dx - 2Af(t_n(k)) \right] \end{aligned}$$

For each k , define $g_k(x) = f(x + t_n(k))$. Since f has a bounded second derivative, so does g_k , and $\|g_k''\| \leq \|f''\|_\infty$. Applying the result from part (a) to $g_k(x)$ with $A = \frac{b-a}{2n}$, we get:

$$\left| \int_{-A}^A g_k(x) \, dx - 2Ag_k(0) \right| \leq \frac{A^3}{3} \|g_k''\| \leq \frac{1}{3} \left(\frac{b-a}{2n} \right)^3 \|f''\|_\infty$$

Summing this bound over all $k = 1, \dots, n$ using the triangle inequality yields:

$$\begin{aligned} \left| \int_a^b f(x) \, dx - \frac{b-a}{n} \sum_{k=1}^n f(t_n(k)) \right| &\leq \sum_{k=1}^n \left| \int_{-A}^A f(x + t_n(k)) \, dx - 2Af(t_n(k)) \right| \\ &\leq \sum_{k=1}^n \frac{1}{3} \left(\frac{b-a}{2n} \right)^3 \|f''\|_\infty \\ &= n \cdot \frac{1}{3} \frac{(b-a)^3}{8n^3} \|f''\|_\infty \\ &= \frac{(b-a)^3 \|f''\|_\infty}{24} \cdot \frac{1}{n^2} \end{aligned}$$

Thus, there exists a constant $C = \frac{(b-a)^3 \|f''\|_\infty}{24}$ dependent only on a , b , and $\|f''\|$ such that the inequality holds.

Problem 3

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{e - \left(1 + \frac{1}{n}\right)^n}{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}}$$

is convergent or divergent.

Solution.

Problem 4 : Basic Exam, Fall 2022

Suppose that $f : [0, 1] \rightarrow \mathbb{R}$ satisfies

$$\sum_{j=1}^n |f(t_j) - f(t_{j-1})|^2 < 100$$

for every choice of $n \in \mathbb{N}$ and of $0 \leq t_0 < t_1 < \dots < t_n \leq 1$. Show that f is Riemann integrable over the interval $[0, 1]$.

Solution.

1. f is bounded in $[0, 1]$: Take $\{0, x, 1\}$.

$$|f(x) - f(0)|^2 + |f(1) - f(x)|^2 < 100 \Rightarrow |f(x) - f(0)|^2 < 100 \Rightarrow -10 < f(x) - f(0) < 10$$

Thus $\forall x \in [0, 1]$ we have $-10 + f(0) < f(x) < 10 + f(0)$.

2. **Number of discontinuities of f :** $\forall \varepsilon > 0$. Assume the contrary that f has more than $\frac{100}{\varepsilon^2}$ jumps of size greater than $\frac{\varepsilon}{2}$, ie there are $\frac{100}{\varepsilon^2}$ points x where $\delta_x > 0$ small enough such that $y \in (x - \delta_x, x)$ and $z \in (x, x + \delta_x)$ such that $|f(y) - f(z)| > \varepsilon$. Take $m = \lceil \frac{100}{\varepsilon^2} \rceil + 1$ such points $x_1 < \dots < x_m$. Let $s_i \in (x_i - \delta_{x_i}, x_i)$ and $t_i \in (x_i, x_i + \delta)$. Then we have the following:

$$\begin{aligned} |f(0) - f(s_1)|^2 + \sum_{i=1}^m |f(s_i) - f(t_i)|^2 + \sum_{i=1}^{m-1} |f(t_i) - f(s_{i+1})|^2 + |f(t_m) - f(1)|^2 &\geq \sum_{i=1}^m |f(s_i) - f(t_i)|^2 \\ &\geq \frac{100}{\varepsilon^2} \varepsilon^2 = 100. \end{aligned}$$

Thus we have a contradiction. Thus there are only a countable number of discontinuities. Hence by Lebesgue criterion f is Riemann integrable.

Problem 5

Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is monotone increasing. Show that f is Riemann integrable.